Integrating Detailed Features and Global Contexts for Semantic Segmentation in Ultrahigh-Resolution Remote Sensing Images

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***Abstract*— Semantic segmentation of ultrahigh-resolution (UHR) remote sensing images is a fundamental task for many downstream applications. Achieving precise pixel-level classification is paramount for obtaining exceptional segmentation results. This challenge becomes even more complex due to the need to address intricate segmentation boundaries and accurately delineate small objects within the remote sensing imagery. To meet these demands effectively, it is critical to integrate two crucial components: global contextual information and spatial detail feature information. In response to this imperative, the multilevel context-aware segmentation network (MCSNet) emerges as a promising solution. MCSNet is engineered to not only model the overarching global context but also extract intricate spatial detail features, thereby optimizing segmentation outcomes. The strength of MCSNet lies in its two pivotal modules, the spatial detail feature extraction (SDFE) module and the refined multiscale feature fusion (RMFF) module. Moreover, to further harness the potential of MCSNet, a multitask learning approach is employed. This approach integrates boundary detection and semantic segmentation, ensuring that the network is well-rounded in its segmentation capabilities. The efficacy of MCSNet is rigorously demonstrated through comprehensive experiments conducted on two established international society for photogrammetry and remote sensing (ISPRS) 2-D semantic labeling datasets: Potsdam and Vaihingen. These experiments unequivocally establish MCSNet stands as a pioneering solution, that delivers state-of-the-art performance, as evidenced by its outstanding mean intersection over union (mIoU) and mean $F1$-score (mF1) metrics. The code is available at: https://github.com/WUTCM-Lab/MCSNet.**

***Index Terms*— Cascade, multilevel fusion, multitask learning, remote sensing, semantic segmentation.**

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I. Introduction

THE accessibility of ultrahigh-resolution (UHR) remote sensing images has greatly increased due to advancements in optical sensors, unmanned aerial vehicles, and satellite technology [1]. These UHR images are used in wide-ranging applications, including disaster assessment, environmental protection, urban planning, transportation routing, and military safety [2], [3]. UHR remote sensing images offer exceptional attributes, such as superior spatial resolution, crystal-clear clarity, timely relevance, and rich information content. However, manually processing these images with expert intervention can be prohibitively expensive, underscoring the importance of optimizing image processing workflows [4]. Consequently, there has been a growing focus on automating the extraction of vital information from UHR remote sensing images in recent years. Semantic segmentation of UHR remote sensing images is a critical step in the overall image processing workflow, and the quality of

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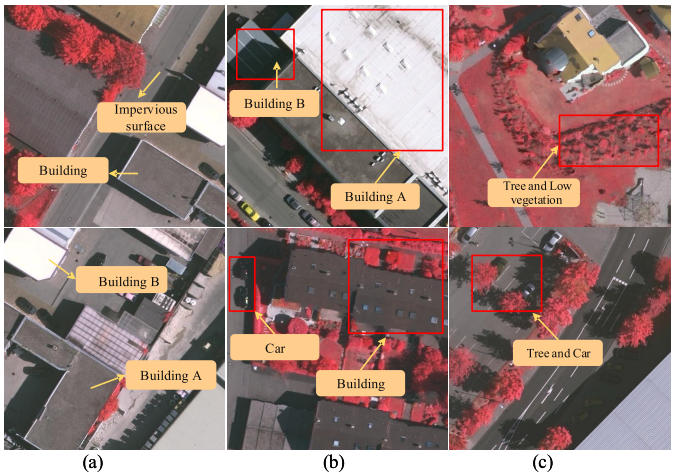

Fig. 1. Examples of features in UHR remote sensing images taken from the ISPRS Vaihingen dataset. (a) Buildings and impervious surfaces are made of the same material, and two different buildings, Building A and Building B are of the same class, but the appearance of the features is completely different. (b) Different buildings, building and car, with huge differences in size. (c) Trees and low vegetation are difficult to distinguish, boundaries are complex, and cars are obscured by trees.

segmentation results has a profound impact on the accuracy of subsequent analysis and processing applied to these remote sensing images.

Recent years have witnessed significant advancements in convolutional neural networks (CNNs), which have provided a robust technical foundation for semantic segmentation. One seminal work that has harnessed the power of deep learning techniques for image semantic segmentation is full convolutional network (FCN) [5]. As a result, numerous pioneering studies have emerged in this field. One notable approach is SegNet, which introduces encoder–decoder architecture to enhance image semantic segmentation [6]. Another innovative architecture, known as U-Net, employs a symmetrical U-shaped structure and incorporates skip connections to fuse features from the encoder with those from the corresponding decoder [7]. Within the DeepLab series [8], atrous convolutions have been integrated into deep CNNs, effectively expanding the receptive field without significantly increasing the network's parameter count. These developments in CNNs have significantly improved the state of the art in semantic segmentation for image analysis.

Semantic segmentation faces distinctive challenges when applied to UHR remote sensing images, as depicted in Fig. 1. These challenges primarily stem from two critical aspects: high interclass similarity and high intraclass dissimilarity. This is especially apparent when comparing distinct buildings within Fig. 1(a), which, despite sharing the same class, exhibit stark differences in terms of color, material, texture, and other visual attributes. Moreover, UHR remote sensing images introduce a higher density of pixel information in feature representations, resulting in increased spatial separation between adjacent categories. This spatial complexity is further compounded by substantial size variations among objects in UHR remote sensing images, both across different classes and within the same class, as exemplified in Fig. 1(b) and (c). Adding to these challenges, object boundaries in UHR remote sensing images often manifest intricate patterns and are prone to significant occlusions, leading to inherent semantic ambiguity. Addressing these complexities effectively requires a holistic approach that combines global contextual information with fine-grained spatial details as guiding cues.

In spite of their contributions and benefits to image analysis, it is essential to acknowledge that conventional CNNs have inherent limitations, especially regarding the size of convolutional kernels that restrict their ability to capture extensive global context information. In addition, the repeated use of pooling operations in CNNs can result in the loss of critical information, exacerbating the challenge of precise pixel classification. The recent breakthroughs achieved by the Transformer architecture, as introduced by Vaswani et al. [9], have opened up new research avenues for modeling global contextual information. The Transformer model is grounded in self-attention mechanisms, offering remarkable capabilities in various domains. Dosovitskiy et al. [10] introduced the Vision Transformer (ViT), showcasing the adaptability of the Transformer structure in image recognition tasks. This marked a significant milestone. Building upon this progress, Liu et al. [11] proposed the Swin Transformer, a model that demonstrated exceptional potential in dense prediction tasks. However, the Swin Transformer's use of sliding-window-based attention, while effective in reducing computational costs, posed a limitation in terms of expanding its receptive field, thereby constraining its ability to capture extensive global context information. Addressing this challenge, Dong et al. [12] recently introduced the CSWin Transformer. This novel approach incorporates a crisscross-style attention mechanism, strategically designed to strike a harmonious balance between the computational efficiency and the capacity to adeptly model global context. The CSWin Transformer method holds particular promise for demanding prediction tasks, including but not limited to target detection and semantic segmentation, where capturing comprehensive global context information are essential for achieving state-of-the-art results. However, Transformer models lack spatial information modeling and exhibit weak capabilities in handling details.

Consequently, a central concern emerges: How can we adeptly strike a balance between leveraging deep global context information and preserving essential shallow spatial detail features when dealing with UHR remote sensing image segmentation?

In this article, we develop a novel semantic segmentation model for UHR remote sensing images, termed multilevel context-aware segmentation network (MCSNet), to finely segment UHR remote sensing images by adequate balanced extraction of different levels of semantic features, as shown in Fig. 2. We first devise a spatial detail feature extraction (SDFE) module that enhances the spatial detail information extraction capability of the encoder. Furthermore, we propose a shared cascaded multiscale fusion (SCMF) module and a dual-branch channel attention (DBCA) module, which collectively constitutes the refined multiscale feature fusion (RMFF) module. The RMFF module refines multiscale feature fusion, enhancing essential but challenging-to-extract information while suppressing redundant and less informative features.

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This promotes improved feature quality and discriminative power in the model.

This study makes the following key contributions.

1) We design MCSNet, a network specifically designed for semantic segmentation tasks. MCSNet integrates the extraction of global contextual and spatial detail information effectively while performing RMFF is performed.
2) We introduce the SDFE module, bolstering the encoder's proficiency in capturing intricate spatial detail information from UHR remote sensing images.
3) We design the RMFF module, comprising the SCMF module and the DBCA module. It excels in integrating encoder-generated feature maps across scales, enabling a thorough fusion of deep global context and shallow spatial details.
4) We conduct a comprehensive comparative analysis of several state-of-the-art methods. In addition, we visualize and analyze the output feature maps of these methods to gain further insights into their effectiveness.

The subsequent sections of this article are organized as follows. Section II provides an overview of the related works relevant to the subject matter. In Section III, a detailed description of the network is presented. Section IV showcases the experimental results, which include comparisons with other methods, thereby demonstrating the effectiveness of our proposed network. Finally, Section V offers a conclusive summary of the findings and contributions of this article.

II. Related Works

A. Vision Transformer

The Transformer architecture made its debut in the domain of machine translation, as showcased in the seminal work of Vaswani et al. [9]. In recent research, the Transformer architecture has garnered significant attention for its applicability to computer vision tasks. Notably, its ability to capture long-range contextual information has made it an attractive choice for demanding tasks such as semantic segmentation. This development in computer vision was initiated by Dosovitskiy et al. [10], who introduced the ViT.

Despite its strengths, ViT encounters substantial challenges when applied to intensive prediction tasks. To mitigate the computational resource consumption of the Transformer and demonstrate its adaptability in the visual domain, Liu et al. [11] propose the Swin Transformer with hierarchical relationships. Swin Transformer adopts a hierarchical feature map generation by splitting the image into smaller blocks. It further performs merging operations on neighboring blocks and extracts deep features using the window-based self-attention mechanism within the deep network. The core innovation of the Swin Transformer lies in the implementation of the shifted window operation, which enables cross-window attention interactions among small blocks. As a result, the Swin Transformer achieves a linear increase in computational complexity with the image resolution, significantly reducing the computational cost.

However, the Swin Transformer is limited by its sliding window approach, resulting in a slow expansion of the receptive field. Achieving approximate global self-attention requires a large number of stacked modules. In response to this constraint, Dong et al. [12] propose a novel visual Transformer backbone with a cross-shaped window. This configuration allows for concurrent self-attention calculation in both horizontal and vertical segments. Each stripe results from segmenting input features into uniformly wide sections. This expands the self-attention calculations within each Transformer block, without introducing additional computational overhead.

B. Multilevel Features Ensembling

In the field of semantic segmentation for UHR remote sensing images, recent studies have predominantly focused on enhancing semantic segmentation performance through the integration of multilevel features. Building upon the U-shaped architecture's encoding and decoding framework, Zhang et al. [13] aim to integrate richer semantic details into shallow-level features. They also seek to infuse improved spatial information into deep-level features. Lin et al. [14] adopt a top–down recursive refinement approach to enhance low-resolution semantic features. The use of a pyramid structure has been widely adopted due to its effectiveness in exploiting multiscale features. Lin et al. [15] devise a multiscale CNN featuring pyramid pooling to capture contextual information from the background of the image patches. The multiscale CNN consists of an image pyramid structure with multibranch multiscale images as network inputs. However, the presence of the multibranch structure leads to significant computational overhead. Addressing this concern, Zhao et al. [16] introduce a feature pyramid ensemble module that integrates multiscale context through the combination of features from various scales. Deeplabv2 and its variants, Deeplabv3 and Deeplabv3+ [8], introduce an extensive convolution alongside the atrous spatial pyramid pooling (ASPP) module. This module employs null convolutions with various expansions in a parallel manner to incorporate multiscale features. Li et al. [17] performs feature extraction and fusion of patches of different resolutions of the same image in image space. However, these methods pay extra attention to detailed information that deep neural networks tend to lose.

C. Multitask Learning for Boundary Detection and Semantic Segmentation

Boundary detection plays a foundational role in various computer vision applications. Numerous research efforts have focused on extracting low-level boundary features using boundary detection operators. For instance, Canny [18] and Heath et al. [19] propose first-order boundary detection operators that leverage the first-order derivatives of an image to identify significant changes. However, the first-order derivatives may generate saddle points, leading to the use of the Gaussian Laplacian (LoG) [19] operator, which captures boundaries by computing second-order derivatives. In recent years, several researchers have improved boundary detection

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performance through CNNs. Holistically-nested edge detection (HED) [20] employs a hierarchical boundary representation and incorporates boundary supervision to guide the learning process. CASENet [21], on the other hand, views boundary detection as a complex multilabel task. In this framework, each boundary pixel is associated with several labels due to its potential involvement in contours or intersections of two or more semantic classes. These boundary details play a crucial role in generating intricate boundaries within semantic segmentation tasks.

Multitask learning, as an approach that enhances model generalization and robustness, facilitates the sharing of representations between primary and secondary tasks [22]. Nonneural algorithms [23], [24], [25] primarily employ paradigm regularization to construct sparse (SP) models and establish relationships among different tasks. Modern CNN-based methods often utilize parameter sharing, encompassing both rigid and flexible parameter-sharing strategies. In the context of rigid parameter sharing, hidden layers are shared across all tasks, while task-specific layers are preserved, resulting in a notable reduction in the likelihood of overfitting. Flexible parameter sharing permits individual tasks to possess their distinct parameters and models, while ensuring normalization across tasks. For example, ResUnet-a [26] regards boundary extraction as a multitask segmentation, projecting multiple labels for each pixel, but more context is needed to understand the significance of this approach. Long et al. [27] introduce a dedicated network layer tailored to the task, serving as prior knowledge to facilitate the learning of intertask relationships. In comparison with the static architecture in the work of Long et al. [27], Lu et al. [28] dynamically group similar models by progressively augmenting the model complexity from a basic configuration. Nevertheless, this method neglects the intertask correlations and might not yield an optimal overall structure.

In recent years, researchers have increasingly utilized boundary detection in joint learning approaches to assist in semantic segmentation tasks, leading to improved semantic segmentation results. For instance, Li et al. [29] introduce a boundary-focused attention mechanism that prioritizes the capture of boundary details alongside semantic information extraction. Zhang et al. [30] present a fusion of the Transformer and CNN architecture within a unified framework, aiming to collectively address the learning objectives of boundary detection and semantic segmentation.

III. Proposed Method

In this section, we provide an overview of our proposed MCSNet, outlining its overall structure and flow. Section III-A provides a comprehensive description of MCSNet's design. Next, in Section III-B, we delve into the working principles and network architecture of the encoder, as well as the design principles underlying the SDFE module. Finally, in Section III-C, we present the structures and principles of the RMFF module.

A. Overall Structure

Our proposed MCSNet follows an encoder–decoder paradigm structure, as illustrated in Fig. 2. Initially, the input raw image X undergoes deep-level global contextual information extraction. In addition, it undergoes shallow-level SDFE. This is achieved through the CSWin Transformer backbone network and three SDFE modules. We adopt a multitask learning approach that involves boundary detection and semantic segmentation to train our network. The encoder produces feature maps that are fed into the boundary detection header, where the boundary detection mask P_{bdr} is generated and supervised learning with the boundary detection loss is applied. Simultaneously, the encoder's feature maps are passed through the decoder to gradually recover their resolution. The SCMF module bridges the encoder and decoder, enhancing deep-level feature maps by complementing them with shallow-level features. This is achieved through a DBCA module at each decoder stage, which enhances features and reduces redundancy. The multiscale feature fusion process described above results in refined feature representations. Then, we concat the decoder's feature map with the feature maps output by the SDFM module and feed it into the semantic segmentation head, where they undergo supervised learning with the semantic segmentation loss, resulting in the final semantic segmentation mask P_{seg}. We finally calculate the weighted sum of L_{seg} and L_{bdr} and train the network.

B. Encoder for Extracting Multilevel Features

Our encoder comprises the CSWin Transformer backbone network, responsible for extracting hierarchically rich deep-level semantic features. In conjunction with this, the SDFE module is employed to capture intricate information from large-size shallow-level feature maps.

1) CSWin Transformer Backbone: The conventional self-attention mechanism demonstrates robust long-range contextual modeling. However, its computational complexity grows quadratically with the feature map dimensions. Consequently, both training and inference costs become significant, especially when working with UHR remote sensing images. To address this, the self-attention mechanism in CSWin Transformer [12] employs a cross-shaped window, conducting parallel self-attention horizontally and vertically. This results in a cross-shaped pattern, enhancing the attention scope and achieving more efficient global self-attention. Therefore, we leveraged CSWin Transformer as the backbone of our network for modeling global contexts.

2) SDFE Module: The SDFE module supports the backbone network in extracting semantic information from spatial details. This allows the network to focus more on details like segmentation boundaries and textures.

Most of the spatial detail information constitutes shallow semantic information, which can be easily lost during the layer-by-layer downsampling process. Therefore, we have designed the SDFE module to effectively capture shallow-level spatial details while avoiding unnecessary parameter increases. The flow of the SDFE module, as depicted in Fig. 2, is outlined as follows:

$$x_1^l = \text{SDFE}_l\left(x_1^{l-1}, y_{l+1}\right) \tag{1}$$

where x_1^l represents the output of SDFE_l, x_1^{l-1} signifies both the input of SDFE_l and the output of SDFE_{l-1}, and Y_{l+1}

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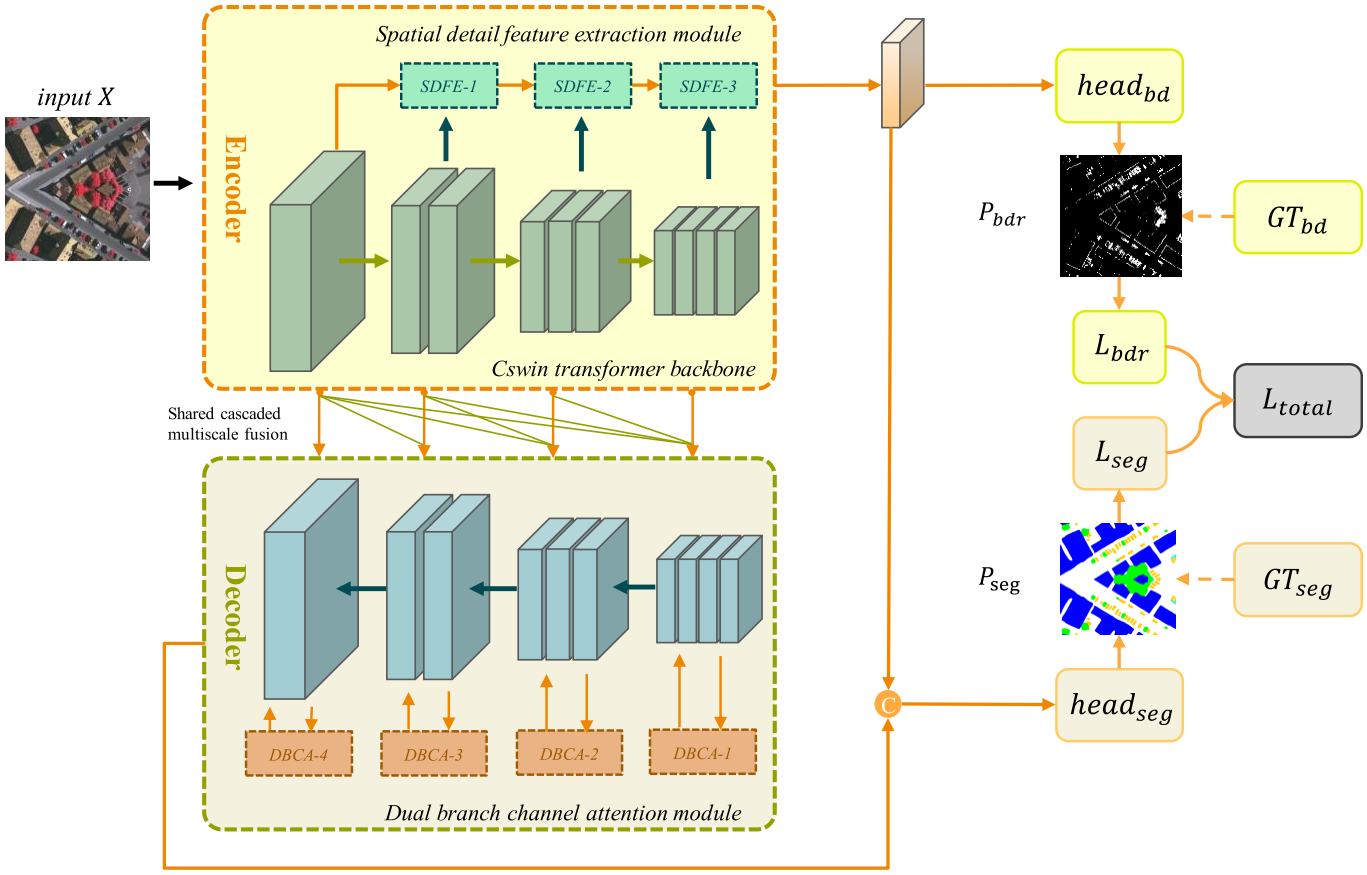

Fig. 2. Pipline of the proposed MCSNet. In this figure, the orange line indicates no processing, the green line indicates downsampling, and the blue line indicates upsampling. head_{bd} and head_{seg} denote the boundary detection head and the semantic segmentation head, respectively, which are used to map the feature maps to the corresponding classification space. L_{bd} and L_{seg} denote the boundary detection loss function and the semantic segmentation loss function for supervised training.

denotes the output of the $(l+1)$th stage of the CSWin Transformer backbone.

Taking the second SDFE module SDFE_2 as an example, its input consists of the output X_1^1 from the previous module SDFE_1 and the output feature map Y_3 from the third layer of the backbone network. SDFE_2 is responsible for extracting shallow-level spatial detail information with the assistance of deep-level semantic information based on the output of SDFE_1. Similarly, SDFE_1 extracts spatial detail semantic information from the output feature map X_1^0 of the first layer of the backbone network, with the assistance of the output feature map Y_2 from the second layer of the backbone network. In other words, whether the module used is SDFE_1, SDFE_2, or SDFE_3, the shallow-level features are extracted from the output X_1^0 of the first layer in the backbone network or from X_1^{l-1} after processing by the guiding module. This extraction is progressively guided by deep-level semantic information.

The specific structure of the SDFE module is depicted in Fig. 3. Assume that S_i represents its shallow-level feature input. For SDFE_1, this input corresponds to the output of the first layer of the backbone network, denoted by X_1^0. In the case of SDFE_2 and SDFE_3, this input takes on the output of the previous SDFE module, denoted as X_1^{l-1}, which shares the same dimensions as the first layer of the backbone network (X_1^0). Now, consider S_j as its deep-level feature input. For

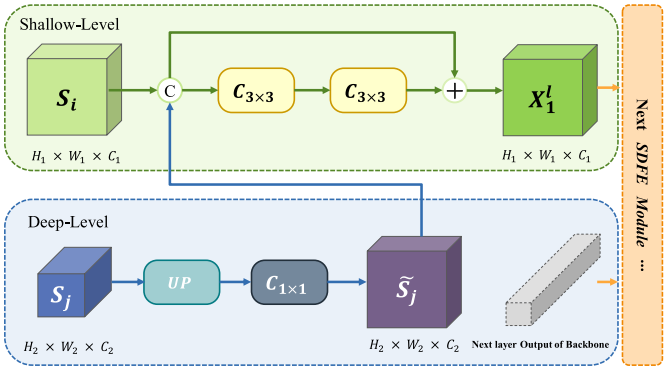

Fig. 3. SDFE module is used to perform the extraction of spatial detail semantic information on the output feature map in the first phase of the backbone.

SDFE_l, this input is derived from the output of the $(l+1)$th layer of the backbone network, represented as Y_{l+1}. Next, we will elaborate on the flow of information using S_i and S_j

$$\widetilde{S}_j = \text{UP}\left(\varphi\left(S_j\right)\right) \tag{2}$$

$$x_1^l = \xi\left(S_i \odot \widetilde{S}_j\right) \oplus S_i. \tag{3}$$

In this description, φ represents a 1×1 convolution operation, and ξ denotes the two consecutive 3×3 convolution opera-

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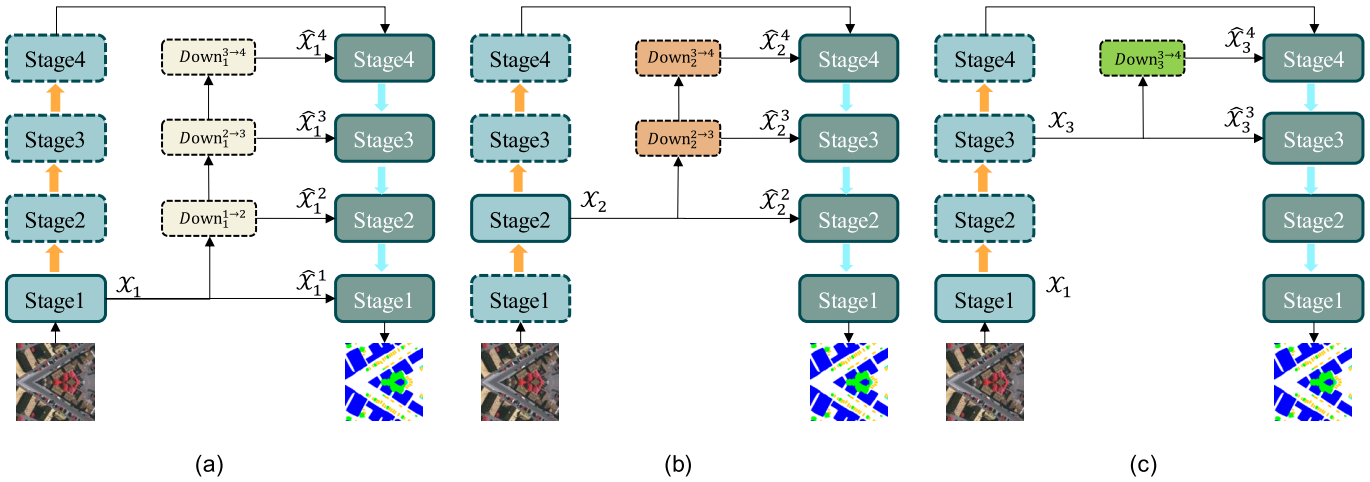

Fig. 4. SCMF module. (a) Three downsampling operations for the output of the first stage of the encoder. (b) Two downsampling operations for the output of the second stage of the encoder. (c) One downsampling operation of the output of the third stage of the encoder.

tions, each followed by batch normalization and the rectified linear units (ReLU) activation function. The process begins by upsampling $S_j \in \mathbb{R}(H_2 \times W_2 \times C_2)$ using a 1×1 convolution to map it to $\widetilde{S}_j \in \mathbb{R}(H_1 \times W_1 \times C_1)$. Subsequently, $\widetilde{S}_j$ is concatenated with S_j and then subjected to two convolution operations, each using a 3×3 convolution kernel with a stride of 1. Batch normalization and the ReLU activation function follow each of these convolution operations. The first convolution operation reduces the number of channels in the feature map by half, changing it from $2 \times C_1$ to C_1, while the second convolution operation maintains the same number of channels as the input. This ensures that the output has the same dimensions and number of channels as the input shallow-level feature map. Consequently, this output can serve as the input for the next SDFE module, continuing the extraction of spatial detail information with the assistance of deep-level features from the output of the subsequent layer in the backbone network.

Our design achieves comprehensive multilevel extraction of shallow-level detailed features aided by deep semantic information. This is accomplished without excessive parameters or computational overhead. Subsequent experiments will validate the accuracy and effectiveness enhancement of our method in semantic segmentation tasks.

C. RMFF Module

For semantic segmentation tasks requiring dense predictions, the necessity for rich deep-level semantic information is evident, but the inclusion of shallow-level spatial structure details is equally important. Transferring local details between feature maps at various scales significantly boosts the network's capabilities. In the prior work, skip connections were typically established between the encoder and decoder at the same scale, enabling communication only at a single scale. This limitation hinders the provision of shallow spatial detail information to deep feature maps. Certainly, attempts have been made to facilitate multiscale communication within a U-shaped architecture for semantic segmentation. However, existing approaches often lead to excessive parameters and computations due to numerous upsampling and downsampling operations. This can result in the inclusion of redundant uninformative features while suppressing crucial features that are challenging to extract. With this issue in mind, we have designed a refined multiscale feature fusion module to enhance the extraction of detailed features that are otherwise difficult to obtain. Below, we provide a detailed description of the structural design and computational principles of the RMFF module.

As shown in Fig. 2, our RMFF module consists of the SCMF module and the DBCA module.

1) SCMF Module: We downsample the large-size low-level feature maps and then fuse the rich spatial detail information from them into the deep semantic features. Take the output $\mathcal{X}_1$ of the first layer of the backbone network as an example. We perform four types of processing on $\mathcal{X}_1$: no operation, one downsampling, two downsamplings, and three downsamplings, as described in the following:

$$\widehat{\mathcal{X}}_1^1 = \mathcal{X}_1 \tag{4}$$

$$\widehat{\mathcal{X}}_1^2 = \mathrm{Down}_1(\mathcal{X}_1) \tag{5}$$

$$\widehat{\mathcal{X}}_1^3 = \mathrm{Down}_2(\mathrm{Down}_1(\mathcal{X}_1)) \tag{6}$$

$$\widehat{\mathcal{X}}_1^4 = \mathrm{Down}_3(\mathrm{Down}_2(\mathrm{Down}_1(\mathcal{X}_1))) \tag{7}$$

where $\widehat{\mathcal{X}}_1^1$, $\widehat{\mathcal{X}}_1^2$, $\widehat{\mathcal{X}}_1^3$, and $\widehat{\mathcal{X}}_1^4$ are the results of $\mathcal{X}_1$ to be aggregated to the output of the first, second, third, and fourth stages of the decoder, respectively. Down_1, Down_2, and Down_3 are the three times of downsampling, and the only difference between them is the number of input and output channels.

We share parameters for different downsampling processes in order to reduce the number of module parameters, as shown in Fig. 4. Down_1 between $\mathcal{X}_1 \Rightarrow \widehat{\mathcal{X}}_1^2$ is also used in $\mathcal{X}_1 \Rightarrow \widehat{\mathcal{X}}_1^3$ and $\mathcal{X}_1 \Rightarrow \widehat{\mathcal{X}}_1^4$, and Down_2 in $\mathcal{X}_1 \Rightarrow \widehat{\mathcal{X}}_1^3$ is also used in $\mathcal{X}_1 \Rightarrow \widehat{\mathcal{X}}_1^4$. Subsequent experiments will verify the effect of parameter sharing on accuracy.

2) DBCA Module: As shown in Fig. 2, we include a DBCA module in the decoder to perform redundant feature suppression and effective feature enhancement on the feature maps after multiscale aggregation.

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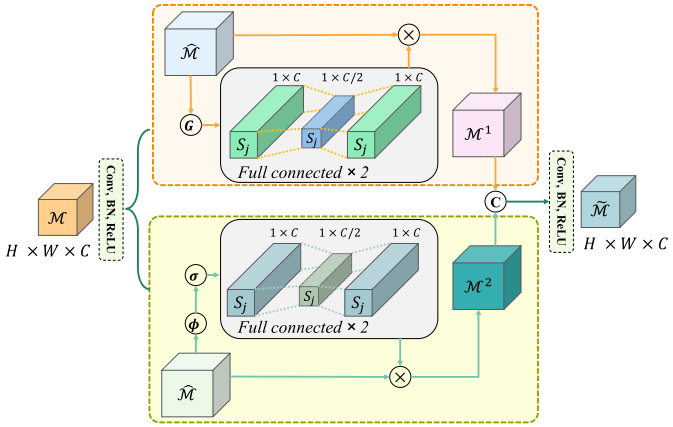

Fig. 5. DBCA module, where one branch is SE channel attention for reducing the effect of redundant features, and the other branch is SP channel attention for enhancing features that are difficult to extract. Here, G denotes global average pooling, ϕ quantifies the sparsity of the feature map, and σ signifies the application of the sigmoid.

As shown in Fig. 5, our DBCA module consists of two branches. One is the standard squeeze-and-excitation (SE) channel attention, and the other is the SP channel attention mechanism based on feature channel sparsity. Next, we describe them in detail.

First, we process the input feature map $\mathcal{M}$ using a convolution with a convolution kernel size of 3×3 and a step size of 1, which is followed by a BatchNorm and an ReLU activation function. The output for the convolution module is denoted by $\widehat{\mathcal{M}}$, which is computed as follows:

$$\widehat{\mathcal{M}} = \text{ConvBN}(\mathcal{M}). \tag{8}$$

After this, SE-Attention and SP-Attention are performed on $\widehat{\mathcal{M}}$ on each of the two branches; the results are spliced together and the number of channels of the spliced results is restored to match the input feature map using $\mathcal{W}^O$

$$\mathcal{M}^1 = \text{SE-Attention}(\widehat{\mathcal{M}}) \tag{9}$$

$$\mathcal{M}^2 = \text{SP-Attention}(\widehat{\mathcal{M}}) \tag{10}$$

$$\tilde{\mathcal{M}} = \text{Concat}\left(\mathcal{M}^1, \mathcal{M}^2\right)\mathcal{W}^o. \tag{11}$$

Next, $\tilde{\mathcal{M}}$ is summed with the input feature map and proceeds to the next stage of the decoder.

SE-Attention is calculated as described in the following equation:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \widehat{\mathcal{M}}(i, j) \tag{12}$$

$$\mathcal{M}^1 = \mathcal{F}_c\left(\mathcal{F}_{c/2}(z_c)\right) \times \widehat{\mathcal{M}} + \widehat{\mathcal{M}}. \tag{13}$$

The vector z_c, whose length is the number of channels C of the feature map $\widehat{\mathcal{M}}$, is obtained by global average pooling of the input feature map $\widehat{\mathcal{M}}$. Subsequently, the length of z_c is transformed to $C/2$ and then reverted back to C using two fully connected layers $\mathcal{F}c/2(\cdot)$ and $\mathcal{F}c(\cdot)$. These layers act as adaptive weights applied to the input feature map $\widehat{\mathcal{M}}$, aiming to selectively suppress redundant features while enhancing those that have a significant impact on the results.

However, we consider that some features may be challenging to extract and require additional attention. This leads to the design of the second part of the DBCA mechanism. Typical categories of images often exhibit strong correlations in channel sparsity. The distribution of SP channels can provide valuable discriminative insights, while feature map sparsity highlights the significance of less frequent features. Therefore, sparsity is obtained by analyzing the properties of nonzero elements in each channel. For the kth feature map $\widehat{\mathcal{M}}^k$ of the input feature map $\widehat{\mathcal{M}}$, sparsity is computed as follows:

$$\Omega_k = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \Delta_{ij} \tag{14}$$

$$\Delta_{ij} = \begin{cases} 1, & \widehat{\mathcal{M}}_{ij}^k > 0 \\ 0, & \widehat{\mathcal{M}}_{ij}^k \leq 0 \end{cases} \tag{15}$$

$$\Gamma_k = 1 - \Omega_k \tag{16}$$

where Γ_k is the sparsity of the kth channel. To emphasize the importance of channels associated with less common features, we redefine the channel weights as follows:

$$\omega_k = \log\left(\frac{C\delta + \sum_{i=1}^{C} \Omega_i}{\delta + \Omega_k}\right) \tag{17}$$

where C represents the number of channels in the feature map and δ is a very small nonzero value, which we set to 0.0001 in our experiments.

We utilize this channel feature sparsity computation to replace the global average pooling component in the SE Block. This allows us to obtain weight vectors that represent feature channel sparsity and apply them to the input feature map.

By utilizing the SCMF module and DBCA module, we achieve precise multiscale feature fusion. This process effectively combines the contextual information from various scales while reducing redundancy and enhancing valuable features, including those that are challenging to extract. Subsequent experiments will demonstrate the effectiveness of this approach.

IV. Experimental Results

A. Experimental Settings

1) Datasets: The Potsdam and Vaihingen datasets for semantic labeling have been made available by the ISPRS committee WG II/4. Each dataset comprises UHR true orthophoto (TOP) tiles and corresponding digital surface models (DSMs) obtained through dense image matching techniques. These datasets include six primary land cover categories: impervious surfaces (such as roads), buildings, low vegetation, trees, cars, and background. In this section, we will introduce two drone image datasets and their associated evaluation procedures.

Vaihingen, located in the southern region of Germany, is predominantly characterized by an urban landscape consisting of neighboring buildings and small-scale multistory structures. The town experiences lower traffic levels. The Vaihingen semantic annotation dataset includes 33 aerial images, each with dimensions of approximately 2100×2100 pixels, captured in infrared-red–green (IR-RG) channels. Each image is accompanied by an associated DSM at a consistent spatial

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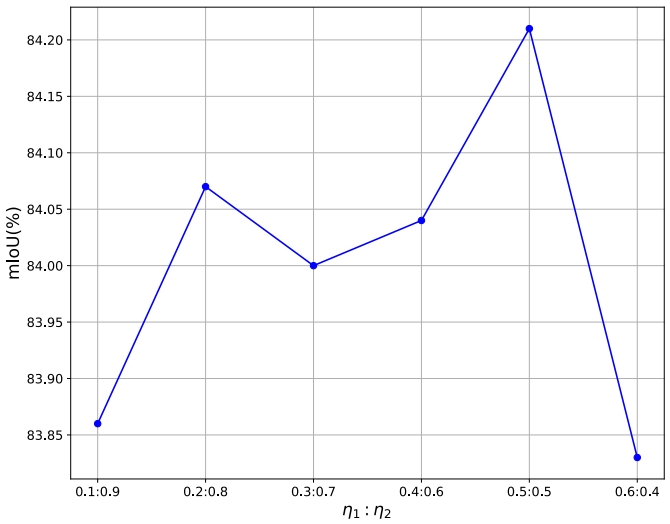

Fig. 6. Effect of multitask loss weights on training results.

resolution of 9 cm. For training purposes, 16 images are provided with ground truth labels, while the remaining 17 images are reserved for testing.

Potsdam, a quintessential historical city located in northeastern Germany, is known for its towering structures, narrow streets, and bustling traffic. The Potsdam dataset is captured from ground level, maintaining a distance of 5 cm. It comprises a total of 38 images, each with dimensions of 6000 × 6000 pixels. The dataset is annotated across six distinct object categories: buildings, cars, low vegetation, impervious surfaces, trees, and background. For training purposes, 24 images are used, leaving 14 images for testing.

2) Evaluation Metrics: For both datasets, we compute validation measures, including intersection over union (IoU) and $F1$ score for each category. We also calculate the overall mean IoU (mIoU) and weighted mIoU to evaluate the results of semantic segmentation. IoU scores are standardized performance metrics employed in object category segmentation tasks. Within a set of images, the IoU metric gauges the likeness between the predicted object regions and the actual regions outlined by ground-truth annotations. The $F1$ score provides a combined assessment of precision and recall for a given class. The equations for these two evaluation metrics are provided as follows:

$$\text{precision} = \frac{\text{TP}}{\text{FP+TP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{18}$$

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{19}$$

$$\text{IoU} = \frac{\text{TP}}{\text{FP+TP+FN}}. \tag{20}$$

In the formula, TPs represents the true positives, FPs signifies false positives, and FNs denotes false negatives. In the computation, TP refers to the count of pixels where both the prediction and actual conditions are accurate. FP corresponds to the count of pixels where the prediction is accurate but the actual condition is inaccurate. FN signifies the count of pixels where the prediction is inaccurate, while the actual condition is accurate.

3) Implementation Details: For both the training and testing datasets, the original image is divided into segments measuring 1024 × 1024 pixels, with an overlapping region of 512 pixels between adjacent segments. During the training process, we use random cropping to further reduce the image size to 512 × 512 pixels.

We train our model using the adaptive matrix estimation (AdamW) optimizer. The initial learning rate is set at 0.001, with the backbone network's learning rate set to 0.0001. We implement the cosine cyclic learning rate strategy according to the following formula:

$$\text{lr} = \text{lr}_{\min} + \frac{1}{2}(\text{lr}_{\max} - \text{lr}_{\min})\left(1 + \cos\left(\frac{T_{\text{cur}}}{T_i}\pi\right)\right) \tag{21}$$

where we set $\text{lr}_{\min}$ to 0, so $\text{lr}_{\max}$ is set to 0.002, and T_i is set to 2. During the initialization training phase, all experiments begin with pretrained weights to accelerate the model's convergence.

Training of the models took place on a GPU machine featuring an NVIDIA 3090 with 24 GB of RAM. A batch size of 4 was used, and the maximum number of training iterations was set to 70.

B. Hyperparameter Settings

1) Multitask Learning for Semantic Segmentation and Boundary Detection: In this article, we use multitask learning for semantic segmentation and boundary detection during training. We have designed loss functions for supervised learning for each of the two tasks. The loss function formula for the first, semantic segmentation, is shown as follows:

$$L_{\text{seg}} = \text{CELoss}(p, q) + \text{DiceLoss}(p, q) \tag{22}$$

$$\text{CELoss}(p, q) = -\sum_{k=1}^{n}(p_k \times \log(q_k)) \tag{23}$$

$$\text{DiceLoss}(p, q) = 1 - \frac{2|p \cap q|}{|p + q|} \tag{24}$$

where p represents the predicted semantic segmentation graph, q represents the ground truth of semantic segmentation, and n represents the number of categories. Our semantic segmentation loss function consists of CELoss and DiceLoss with a ratio of 1:1.

The loss function formula for the second, boundary detection, is shown as follows:

$$L_{\text{bdr}} = \text{BCELoss}(p, q) + \text{BF1Loss}(p, q) \tag{25}$$

$$\text{BCELoss}(p, q) = -(q \log(p) + (1 - q) \log(1 - p)) \tag{26}$$

$$\text{BF1Loss} = 1 - 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{27}$$

where p represents the predicted boundary detection graph and q represents the ground truth for boundary detection. Precision is the precision rate, defined as precision = TP/(TP + FP), and recall is the recall rate, defined as recall = TP/(TP + FN). Our boundary detection loss function consists of BCELoss and BF1Loss in a 1:1 ratio.

Our training network uses both semantic segmentation and boundary detection loss functions

$$L_{\text{total}} = \eta_1 L_{\text{seg}} + \eta_2 L_{\text{bdr}} \tag{28}$$

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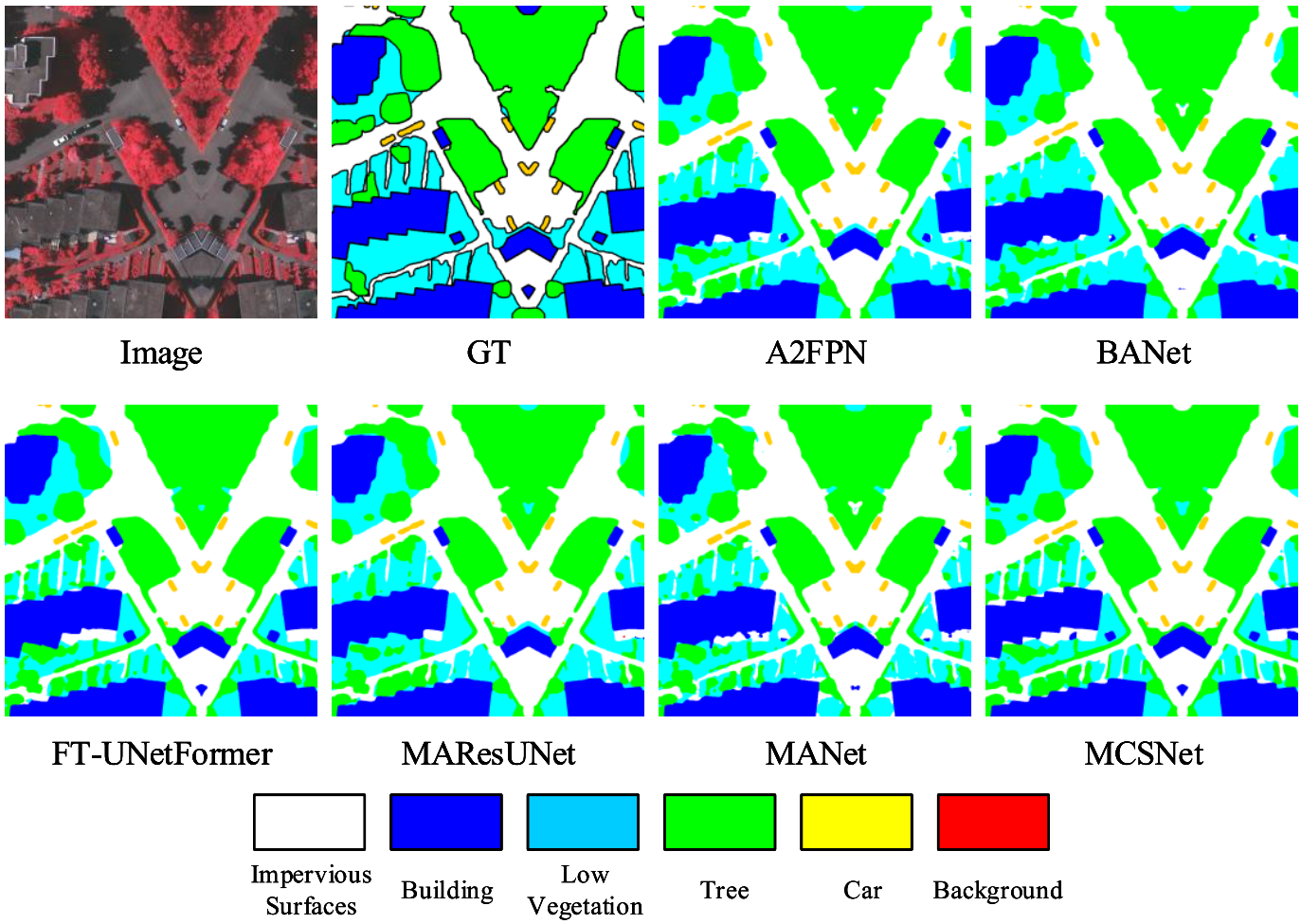

Fig. 7. Comparison of state-of-the-art methods on the Vaihingen test set.

TABLE I
MIOU AND PARAMETER SIZES OF BACKBONE NETWORKS OF DIFFERENT SIZES—VAIHINGEN DATASET

Backbone	Vaihingen		Potsdam	
	mIoU (%)	Param (M)	mIoU (%)	Param (M)
CSWin-tiny	84.26	26	86.92	26
CSWin-small	**84.31**	39	**87.36**	39

TABLE II
NUMBER OF CONVOLUTIONS IN THE SDFE MODULE WAS TESTED ON THE VAIHINGEN DATASET

Num of Conv	mIoU (%)	Params (M)
1	83.99	0.53
2	**84.26**	0.92
3	84.01	1.42
4	84.13	1.93

TABLE III
MULTISCALE FUSION APPROACHES TESTED ON THE VAIHINGEN DATASET

Multi-scale feature fusion approach	mIoU (%)	Params (M)
Non-cascading	83.92	14.17
Non-shared cascade fusion	84.01	20.40
Global shared cascade fusion	83.94	6.75
Locally shared cascade fusion	**84.26**	17.43

in which

$$\eta_1 + \eta_2 = 1. \tag{29}$$

To assess the impact of the two tasks on overall semantic segmentation outcomes, we conducted experiments on the Vaihingen dataset; the findings are presented in Fig. 6. Notably, we observed that setting both η_1 and η_2 to 0.5 yielded the most favorable results, indicating the critical role of spatial detail information in the semantic segmentation task. However, tasks with higher loss weights tended to dominate the backpropagation process during learning. This dominance could lead the network to over emphasize one task at the expense of the other. Therefore, we selected 0.5 as the optimal values of η_1 and η_2 for all our experiments.

2) Backbone Settings: Table I records the segmentation accuracies of our networks with different backbone networks, CSWin-tiny and CSWin-small, on the Vaihingen dataset. It is worth noting that CSWin-small is deeper, allowing for the extraction of more semantic information, making it a better choice than CSWin-tiny.

3) Module Settings: In our network, we designed the SDFE module to complement the backbone network with a substantial amount of spatial detail information, including boundary

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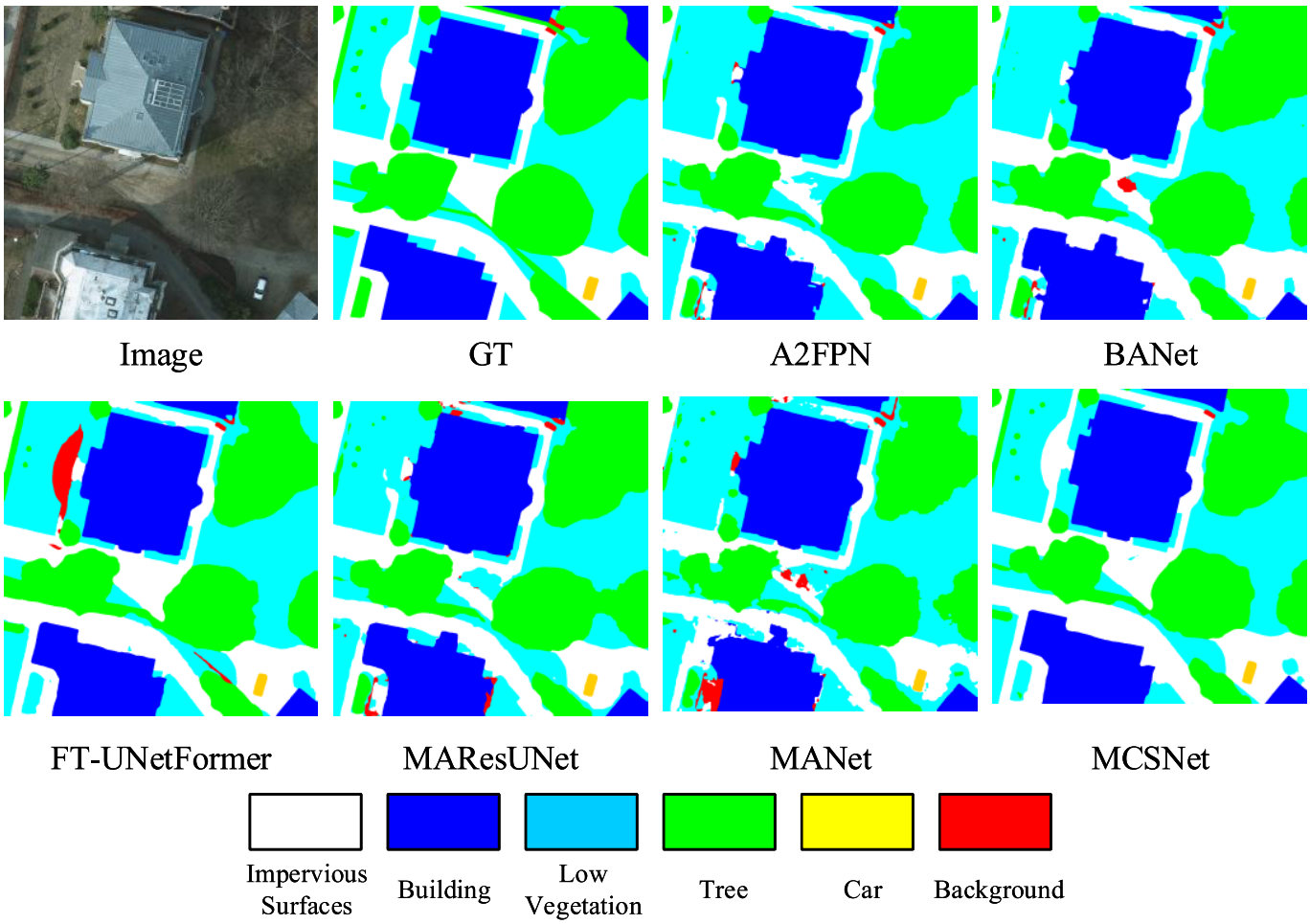

Fig. 8. Comparison of state-of-the-art methods on the Potsdam test set.

TABLE IV
RESULTS OF QUANTITATIVE COMPARISON EXPERIMENTS ON THE VAIHINGEN DATASET

Method	Imp. Surf.	Building	Low veg	Tree	Car	mF1 (%)	mIoU (%)
FCN [5]	89.73	93.17	80.57	88.89	71.55	84.78	73.45
DANet [31]	91.38	94.10	83.09	89.02	76.79	86.88	77.32
PSPNet [16]	91.38	94.20	83.05	88.71	75.02	86.47	76.78
DeepLab V3+ [8]	91.63	94.09	82.51	87.99	77.66	86.77	77.13
EaNet [32]	91.71	94.86	84.23	90.06	82.04	88.58	79.83
BotNet [33]	92.22	94.48	83.99	89.57	82.93	88.63	79.89
ResT [34]	92.46	95.16	83.72	89.51	84.27	89.03	80.52
MANet [35]	93.02	95.47	84.64	89.98	88.95	90.41	82.71
MAResUNet [36]	93.27	95.75	85.60	90.79	89.18	90.92	83.53
BANet [37]	92.81	95.35	85.35	90.87	88.48	90.57	82.95
FT-UNetFormer [38]	93.40	95.91	85.15	90.68	90.72	91.17	83.97
A^2FPN [39]	93.03	95.61	85.37	90.84	89.23	90.92	83.36
DCSwin [40]	93.11	95.85	84.97	90.77	89.19	90.78	83.32
CMTFNet [41]	90.61	94.21	81.93	87.56	82.77	87.42	77.95
GPINet [42]	93.90	95.98	86.69	92.39	89.17	**91.58**	82.57
MCSNet-tiny	93.52	95.85	85.83	91.12	90.48	91.36	84.26
MCSNet-small	93.50	95.89	85.68	91.07	90.77	91.38	**84.31**

information. We conducted experiments on the vaihingen dataset, the results of which are shown in Table II.

When two convolutional modules are used, our SDFE module can extract detailed features from the shallow subfeature

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TABLE V
RESULTS OF QUANTITATIVE COMPARISON EXPERIMENTS ON THE POTSDAM DATASET

Method	Imp. Surf.	Building	Low veg	Tree	Car	mF1 (%)	mIoU (%)
FCN [5]	90.84	95.59	84.10	84.75	84.95	88.05	79.53
DANet [31]	91.94	96.38	86.00	87.67	86.01	89.60	81.40
PSPNet [16]	92.20	96.11	86.94	88.34	86.30	89.98	81.99
DeepLab V3+ [8]	92.09	95.28	85.55	86.54	94.81	90.86	84.24
EaNet [32]	92.87	96.30	86.16	87.99	95.30	91.73	85.34
BotNet [33]	92.34	96.30	87.32	88.74	94.17	91.77	84.97
ResT [34]	91.14	95.11	86.30	87.27	94.63	90.89	83.50
MANet [35]	93.40	96.96	88.32	89.36	96.48	92.90	86.95
MAResUNet [36]	90.30	93.81	86.59	88.67	95.93	91.06	83.76
BANet [37]	92.07	95.52	86.58	88.04	95.49	91.54	84.61
FT-UNetFormer [38]	93.36	97.04	88.32	89.60	96.53	92.98	87.09
A^2FPN [39]	92.74	96.22	86.96	88.09	95.98	92.00	85.42
DCSwin [40]	93.57	97.03	88.09	89.48	96.19	92.87	86.90
CMTFNet [41]	92.12	96.41	86.43	87.26	92.41	90.93	83.57
GPINet [42]	93.78	97.50	88.15	88.86	96.70	93.00	83.33
MCSNet-tiny	93.59	97.09	88.15	89.63	95.98	92.89	86.92
MCSNet-small	93.81	97.31	88.28	89.75	96.51	**93.13**	**87.36**

maps, leading to the best results effectively. However, it is important to note that increasing the number of convolutional layers beyond this point does not improve segmentation performance. This is because adding more convolutions introduces additional gradients during training, making it challenging to achieve better results.

We conducted tests on our SCMF module, as illustrated in Fig. 4. Our SCMF module employs a local shared cascaded fusion approach, where parameters are shared among the downsampling operations within a specific stage of the encoder. In addition, we designed noncascaded fusion, nonshared cascaded fusion, and global shared cascaded fusion. In noncascaded fusion, only one downsampling operation is used between any two stages. The structures of nonshared cascaded fusion and global shared cascaded fusion are similar to our SCMF module. However, in nonshared cascaded fusion, unique weights are assigned to each downsampling operation, while in global shared cascaded fusion, weights are shared across downsampling operations between different stages of the encoder. We evaluated these four design approaches on the vaihingen dataset, presenting the results in Table III. The local shared cascaded fusion used in our SCMF module achieves the best results and does not introduce too many extra parameters.

C. Comparison With State-of-the-Art Methods

1) Validity on the Vaihingen Dataset: We evaluate MCSNet's segmentation performance against various contemporary techniques on the Vaihingen test dataset. Note that the methods considered use identical input and parameter configurations. The results in Table IV show that our network achieves superior performance in terms of mIoU, mean $F1$ (mF1), and $F1$ scores for individual categories.

As can be seen from Table IV, FCN excels in segmenting larger targets such as buildings but struggles with smaller objects like cars. This limitation arises from FCN's relative neglect of intricate details that are challenging to extract. In contrast, DANet introduces a dual attention mechanism capable of efficiently capturing global and local information across multiple scales, resulting in greater accuracy than FCN across all categories. PSPNet and DeepLab V3 utilize pyramid pooling modules to capture multiscale contextual information, contributing to enhanced performance. EaNet employs an extensive kernel pyramid pooling module, capturing diverse multiscale contexts and continuous feature relationships effectively, thereby boosting overall accuracy. BoTNet and ResT, based on Transformer networks, excel at capturing long-range contexts through attention mechanisms. MANet and MAResUNet introduce a linear attention mechanism, extracting context dependencies through multiple efficient attention modules. BANet employs a bidirectional

TABLE VI
COMPARISON OF THE COMPLEXITIES OF OUR METHOD WITH OTHERS

Methods	Param (M)	mIoU (%)
FCN	32.95	73.45
DANet	49.51	77.32
PSPNet	51.43	76.78
DeepLab V3+	39.63	77.13
EaNet	83.57	79.83
MANet	35.86	82.71
MAResUNet	28.49	83.53
BANet	12.72	82.95
FT-UNetFormer	95.97	83.97
A^2FPN	24.26	83.36
DCSwin	32.95	83.32
CMTFNet	30.07	77.95
GPINet	27.70	82.57
MCSNet-tiny	42.41	84.26
MCSNet-small	54.74	**84.31**

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TABLE VII
VALIDATION OF SDEM AND RMFF MODULES FOR SEGMENTATION MIOU (%) ENHANCEMENT ON VAIHINGEN AND POTSDAM DATASETS

Method	SDFE	RMFF (SCMF)	RMFF (SCMF+DBCA)	Vaihingen	Potsdam	Params (M)
Baseline	-	-	-	82.82	85.16	23.47
MSCNet-1	✓	-	-	83.32	85.78	24.39
MCSNet-2	✓	✓	-	83.83	86.49	41.82
MCSNet-3	✓	✓	✓	**84.26**	**86.92**	42.41

aggregation module, enhancing feature representation by transferring information in both forward and backward directions. FT-UNetFormer integrates a multiscale feature fusion mechanism and employs deep separable convolutions to enhance both model performance and efficiency. DCSwin leverages deep channelwise and spatialwise attention mechanisms to enhance feature representation across various computer vision tasks. Finally, A2-FPN introduces an attention aggregation mechanism, further improving feature expression capability. CMFTNet and GPINet attempt to fuse local and global information.

Our MCSNet extracts varying levels of feature information through the backbone and spatial detail information extraction module of the CSWin Transformer. It then aggregates these diverse features via the refined multiscale fusion module. As a result, our approach enhances both the segmentation continuity of large targets and the segmentation accuracy of small targets. For instance, we observe improved accuracy in segmenting large targets such as concrete, ground, and buildings, as well as improved accuracy in segmenting cars. In addition, MSCNet handles the details of segmentation texture and segmentation boundary better. For example, its segmentation accuracy on trees and low vegetation, which have complicated segmentation boundaries, has been improved to a certain extent.

2) Validity on the Potsdam Dataset: We also compared the segmentation performance of MCSNet with these state-of-the-art methods on the Potsdam test set. The results are documented in Table V. As with the Vaihingen dataset, MCSNet demonstrates superior performance in terms of mF1 and mIoU compared with the other networks. Notably, most of the comparison methods yielded better outcomes on the Potsdam dataset than on the Vaihingen dataset, likely due to the larger training sample size available in the Potsdam dataset.

Figs. 7 and 8 show the image visual segmentation results of MCSNet and some state-of-the-art methods on the Vaihingen and Potsdam test sets, respectively. For other methods, the results are poor in some details such as the segmentation boundary, and problems such as boundary jaggedness and boundary-blurring occur. As shown in Fig. 8, A2FPN, BANet, FT-UNetFormer, and MAResUNet have boundary blurring in the processing of complex building boundaries. MANet, on the other hand, has more severe boundary jaggedness. Our MCSNet achieves finer segmentation and improves the processing of segmentation details.

3) Model Sizes: Table VI displays the parameter sizes of methods within the same operating environment. Upon reviewing Table VI, it becomes evident that MCSNet achieves superior results without introducing an excessive number of parameters.

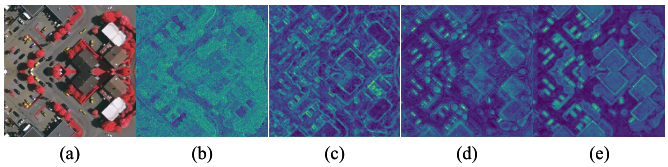

Fig. 9. Feature maps after processing by SDFE modules. (a) Original image, (b) output of the first stage of the encoder (the input of the first SDFE module), (c) output of the SDFE module (the input of the second SDFE module), (d) output of the second SDFE module (the input of the third SDFE module), and (e) output of the third spatial detail information extraction module.

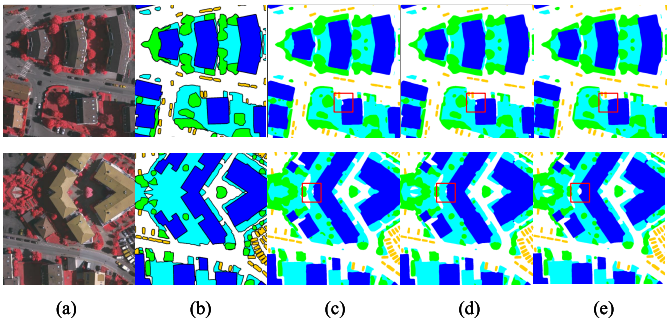

Fig. 10. Effect of the RMFF module on the segmentation results. (a) Original input image, (b) ground truth, (c) output of the network without the RMFF module, (d) output of the network with only the SCMF module, and (e) output of the network with the SCMF module and DBCA module.

D. Assessment of Module Effectiveness

1) SDFE Module: To confirm the effectiveness of the SDFE module, we provide visual illustrations in Fig. 9, displaying the outputs of each SDFE module alongside the feature maps of the initial inputs. We can observe that small targets, such as cars, and detailed information like segmentation boundaries and textures, gradually become more prominent and receive increased attention as they pass through our SDFE modules. This information complements the global context extracted by the backbone network, contributing to the overall land cover classification. Based on this analysis, the spatial detail information extraction module proves to be effective.

In addition, we conducted quantitative analysis on the SDFE module, as shown in Table VII. Compared with the baseline, the SDFE module improved the model's segmentation results (mIoU) by 0.50% and 0.62% on the Vaihingen dataset and Potsdam dataset, respectively.

2) RMFF Module: We verified the effectiveness of the RMFF module; Fig. 10 illustrates its impact on the results. Our RMFF module ensures both the continuity of segmentation

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for large targets and the accuracy of segmentation boundaries. It achieves this by adaptively aggregating features at various scales and levels, balancing the seamless segmentation of significant objects with meticulous handling of segmentation details in the image.

Similarly, Table VII also presents the quantitative results of RMFF. On the Vaihingen dataset, RMFF improved the segmentation mIoU by a total of 0.94%. Among which, SCMF contributed 0.51%, and DBCA contributed 0.43%. On the Potsdam dataset, RMFF is improved by 1.14%.

V. Conclusion

In this study, we introduce the MCSNet designed explicitly for the semantic segmentation of UHR remote sensing images. The architecture revolves around an encoder–decoder structure. In the encoder, we introduce the SDFE module to infusion the spatial detail information, which is often overlooked and enrich the encoder's capabilities.

In the decoder phase, we introduce the RMFF module, comprising the SCMF module and the DBCA module. It refines multiscale feature fusion that seamlessly integrates deep-level and shallow-level features.

To further enhance segmentation precision, we employ a multitask learning strategy, simultaneously addressing semantic segmentation and boundary detection during training. This approach mitigates noise and elevates boundary localization accuracy.

Extensive experimentation validates the efficacy of MCSNet on both the Potsdam and Vaihingen datasets. A series of ablation experiments confirm the contributions of these designed modules. In a comprehensive evaluation, MCSNet outperforms state-of-the-art methods on the Vaihingen and Potsdam datasets.

In summary, this article introduces MCSNet as a powerful tool for accurate and comprehensive semantic segmentation of UHR remote sensing images. While achieving favorable segmentation results, we acknowledge the model's computational complexity and the potential for simplifying the network architecture and enhancing SDFE modules in future work.

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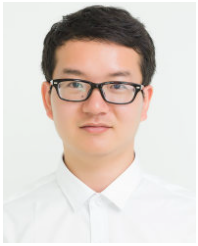

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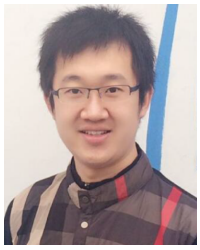

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